

KEY TAKEAWAYS

- 1 In India, the transport sector accounts for 10% of greenhouse gas emissions
- 2 India, the world's third-largest auto market, is expected to add 10 million vehicles by 2030
- 3 In CY2024, global EV sales surpassed 17 million units, pushing EVs' share in the global car market above 20% for the first time
- 4 EV penetration in the domestic market is 8%; India is targeting 30% by 2030
- 5 In FY25, sales of CNG cars surpassed sales of diesel vehicles; CNG accounts for over 15% of the domestic car market

gy Agency (IEA). The agency projects that one in four cars sold in 2025 will be electric, rising to over two in five by 2030. "Countries like China, the US, and EU members are investing heavily in EV infrastructure and battery innovation," says Grant Thornton's Mehra. EV adoption in passenger vehicles in India is 2–3%. But 2024 saw a 27% rise in overall sales of EVs to a record 1,949,114 units, driven by two-wheelers (59%) and three-wheelers (35%).

India is targeting 30% EV penetration by 2030, supported by policies such as 5% GST on EVs, reduced road taxes in several states and subsidies under FAME (Faster Adoption of Manufacturing of Electric Vehicles and Hybrid Vehicles) and PM E-Drive (PM Electric Drive Revolution in Innovative Vehicle Enhancement) schemes. The Scheme for Promotion of Manufacturing of Electric Passenger Cars in India, offering import duty rebates, is also expected to attract global players.

Automakers such as Maruti Suzuki, Tata Motors, Mahindra & Mahindra and Hyundai are already ramping up their EV portfolios. Tata Motors, the country's largest EV manufacturer, has planned an investment of ₹16,000–18,000 crore by 2030.

"If we go by what the government and the Bureau of Energy Efficiency have been saying, it would mean 10%-plus EV penetration for a manufacturer like us. We are at 11%. So, we are comfortable," Shailesh Chandra, Managing Director, Tata Motors Passenger Vehicles and Tata Passenger Electric Mobility, said in a recent analyst call. The company aspires to achieve more than 30% EV penetration by FY30. "We are pretty much on track," he said.

Maruti Suzuki, the country's largest passenger vehicle manufacturer, plans to launch four EVs by 2030. Its first EV will roll out this September. "It is a freshly designed EV on a new platform which offers several safety and performance advantages—500 km range, battery safety, larger cabin space, Level 2 ADAS, seven airbags. Above all, the customer will get a lot of confidence when he sees the same car being exported to advanced markets like Europe and Japan," says Rahul Bharti, Senior Executive Officer, Corporate Affairs, Maruti Suzuki India Ltd. Despite being late to the EV party, Maruti Suzuki is taking a measured approach. "We decided to adopt a more diversified approach and did not want to put all our eggs in one basket," Chairman R.C. Bhargava wrote in the FY24 annual report.

M&M is also treading cautiously. Despite committing ₹12,000 crore to EV arm Mahindra Electric Automobile Ltd (MEAL) and launching electric models like the BE.6 and XEV.E9, it is scaling up slowly. "This is a business in which we don't want to be rash. There is a lot of product complexity. We are learning new technologies as we ramp up, and so are our suppliers. We have to be cautious," Rajesh Jejuri, ED, Auto & Farm, Mahindra & Mahindra, said in a recent analyst call.

But electrification faces hurdles such as affordability, shortage of charging infrastructure and range anxiety. "For two-wheelers and three-wheelers, we will obviously hit it, but for four-wheelers, it is going to be tough. At just 3% penetration in 2025, we don't have a strong base. Achieving 30% in five years is a 10x leap, which is difficult," says Rajat Mahajan, Partner and Automotive Sector Leader at Deloitte India.

Grant Thornton's Mehra concurs. "One of the foremost hurdles is the high upfront cost of electric and hybrid vehicles. Charging infrastructure is underdeveloped in many regions. India has a little over 25,000 public charging stations, concentrated in urban regions, limiting consumer confidence," says Mehra.

HYBRIDS BRIDGE GAP

While it will take some time for Indian customers to widely adopt EVs, hybrids have emerged as a viable in-